

ARTIFACT REFERENCE LIBRARY

AIDLC

Input & Output Artifact Templates & Examples

All 8 AIDLC Phases · EA Artifacts · Phase Gates

Every artifact shown as a live template with field definitions and example content

1

Discovery

2

Feasibility

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Data

4

Architecture

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Development

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Evaluation

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Deployment

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Monitor

EA

Artifacts

ARTIFACTS

40+ Templates

PHASES

8 AIDLC + EA

FORMAT

Field-by-field

HOW TO READ THIS DOCUMENT

Element	Description
Phase Header	Coloured banner for each AIDLC phase with phase number, name, and governance gate
I/O Flow Box	Green left = inputs consumed by this phase. Blue right = artifacts produced by this phase
Artifact Card	Each artifact shown as a template. The coloured badge (INPUT/OUTPUT/EA) indicates artifact role
Field Grid	Gray background = field label. Blue italic text = example value you would fill in for a real project
Table Sections	Structured tables within artifacts (e.g., risk register rows, bias metrics) shown with example data
Section Bars	Dark coloured bars separate major sections within each artifact template

EXAMPLE CONTENT: All field values shown in blue italic are illustrative examples for a hypothetical "Credit Risk AI Scoring Model" use case at a financial services enterprise. Replace all blue italic values with your actual project data.

PHASE
1

Discovery & Ideation

Define the AI use case, business value hypothesis, and initial feasibility signal

GATE

AI Governance Council Registration

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE

- Strategic Business Goals
- Market / Competitive Analysis
- Stakeholder Pain Points
- Regulatory Landscape Brief

OUTPUTS FROM THIS PHASE

- AI Use Case Charter (AUC-001)
- Business Value Canvas (BVC-001)
- Initial Risk Classification Sheet (RCS-001)
- Executive Sponsor Sign-off (ESO-001)

OUTPUT

AUC-001

AI Use Case Charter

Owner: Business Owner / Product Manager | Phase: Phase 1: Discovery

CHARTER IDENTIFICATION

Use Case ID	AUC-2026-FIN-047
Use Case Name	Credit Risk AI Scoring Model — Retail Lending
Business Domain	Retail Banking — Credit Risk Management
Business Owner	Sarah Chen, VP Credit Risk, Retail Banking
Product Manager	David Okafor, Senior PM, AI & Analytics
Date Initiated	12 February 2026
Charter Version	v1.2
Status	Approved — Proceeding to Phase 2

PROBLEM STATEMENT

Problem Statement	The current credit scoring process takes 72 hours on average, relies on manual review for 40% of applications, and results in a 12% false-rejection rate for creditworthy customers in underserved segments. This causes customer churn, missed revenue, and equity concerns.
Business Pain Points	72-hour decision cycle; 40% manual review burden; 12% false-rejection rate; Inability to serve thin-file customers; Manual process inconsistency across regions
Target Opportunity	Reduce decision time to <60 seconds for 80% of applications; reduce false-rejection rate to <5%; enable thin-file credit assessment via alternative data

AI SOLUTION HYPOTHESIS

Proposed AI Solution	ML-based credit scoring model using gradient boosting + alternative data (utility payments, rent history) combined with LLM-powered adverse action explanation generator
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AI Approach	Supervised ML classification (creditworthy / not creditworthy) + RAG-powered explanation generation for regulatory adverse action notices
AI vs Non-AI Decision	AI preferred: decision volume (50,000/month), pattern complexity, regulatory explainability requirement, real-time speed requirement. Non-AI alternatives (rule engine, scorecard) evaluated and rejected.
Build vs Buy vs Fine-Tune	Buy: Foundation credit risk models (FICO, Equifax). Fine-tune: LLM for adverse action explanation. Build: Alternative data pipeline and ensemble orchestration.

ROI & VALUE HYPOTHESIS

Metric	Baseline	AI Target	Annual Value Impact
Decision Cycle Time	72 hours	<60 seconds	£2.1M saved (FTE reduction + faster revenue recognition)
Manual Review Rate	40%	<8%	£1.4M saved (analyst hours)
False Rejection Rate	12%	<5%	£3.8M recovered revenue (re-captured creditworthy customers)
Customer NPS (Credit)	34	>55	Brand value; reduced churn
Total Projected Annual ROI	—	—	£7.3M (payback period: 9 months)

INITIAL RISK FLAGS

Preliminary EU AI Act Tier	HIGH RISK — Annex III Category: Access to financial services (creditworthiness assessment). Full AIDLC + FRIA required.
Protected Attributes Involved	Age, Gender, Ethnicity, Postcode (proxy for protected attributes). Bias mitigation mandatory from Phase 3.
Regulatory Bodies	FCA (UK), PRA, ICO, EU AI Act National Competent Authority (if EU customers)
Explainability Requirement	MANDATORY — ECOA/Regulation B-equivalent (UK) requires adverse action notice with specific reason codes. Every rejection must be explainable.

EXECUTIVE SPONSOR SIGN-OFF

Executive Sponsor	James Hartley, Chief Risk Officer
Sign-off Date	19 February 2026
Approval Conditions	1. Full FRIA completed before development begins. 2. External fairness audit in Phase 6. 3. Quarterly bias monitoring post-deployment.
Escalation Path	AI Governance Council → CRO → Board Risk Committee

OUTPUT

BVC-001

Business Value Canvas

Owner: Product Manager | Phase: Phase 1: Discovery

Canvas Dimension	Example Content
AI System	Credit Risk AI Scoring Model (AUC-2026-FIN-047)
Customer Segments Served	Retail banking applicants: prime, near-prime, thin-file, underserved
Value Propositions	60-second credit decisions; fair thin-file assessment; consistent explainable outcomes; 24/7 availability
Key Activities	Alternative data ingestion, real-time scoring, adverse action generation, bias monitoring
Key Resources	Credit data (bureau + alternative), GPU inference cluster, model team, compliance expertise
Key Partners	Equifax/Experian (bureau data), Utility data providers, FCA regulatory counsel
Revenue Streams	Net interest income on approved loans, reduced cost of risk (fewer defaults), fee income
Cost Structure	Model development (£420K), annual MLOps (£180K), compliance (£95K), data licensing (£240K)
Success Metrics	Decision time <60s (80% of apps); false rejection <5%; bias delta <2%; Explainability coverage 100%
Failure Risks	Regulatory rejection (FRIA outcome); bias discovery post-deployment; data quality failure; model drift

OUTPUT

RCS-001

Initial Risk Classification Sheet

Owner: AI Governance Council | Phase: Phase 1: Discovery

Use Case ID	AUC-2026-FIN-047
EU AI Act Annex III Check	YES — Category 5(b): AI used for creditworthiness assessment of natural persons
Assigned Risk Tier	TIER 2 — HIGH RISK
FRIA Required	YES — Fundamental Rights Impact Assessment mandatory
External Audit Required	YES — Independent conformity assessment before deployment
Governance Intensity	Full AIDLC all 8 phases; AI Governance Council sign-off at Phases 2, 4, 6, 7
Classification Approved By	Priya Sharma, AI Compliance Lead
Classification Date	22 February 2026

PHASE
2

Feasibility & Risk Assessment

Validate technical feasibility, assess full risk exposure, obtain governance approval to proceed

GATE
AI Governance Council Go/No-Go

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE	OUTPUTS FROM THIS PHASE
→ AI Use Case Charter (AUC-001)	→ Feasibility Report (FR-001)
→ Business Value Canvas (BVC-001)	→ AI Risk Register (RR-001)
→ Initial Risk Classification (RCS-001)	→ FRIA (FRIA-001)
	→ TPRM Vendor Assessment (TPRM-001)
	→ Compliance Obligation Matrix (COM-001)

OUTPUT

FR-001

Technical Feasibility Report

Owner: AI Architect + Data Architect | Phase: Phase 2: Feasibility

TECHNICAL FEASIBILITY ASSESSMENT	
Data Availability	Bureau data: HIGH availability (Equifax, Experian APIs). Alternative data (utility): MEDIUM (3 providers identified, contract negotiation required). Thin-file historical: LIMITED — synthetic augmentation required.
Data Volume Assessment	Training set: 2.3M historical applications (36 months). Positive class ratio: 73% approved. Class imbalance strategy: SMOTE + cost-sensitive learning.
Model Approach Selected	XGBoost ensemble (primary scorer) + SHAP explainer (feature importance) + GPT-4o fine-tuned (adverse action narrative generation). Architecture: RAG for regulatory reason code library.
Infrastructure Feasibility	Azure ML available for training. Azure AI Content Safety for LLM outputs. Inference SLA: p99 <200ms achievable with KServe + model caching. GPU budget: £45K/year training, £28K/year inference.
Integration Feasibility	Core banking API (Temenos): REST available, 200ms SLA. Bureau APIs: SOAP to REST adapter required (2 weeks build). Event streaming: Azure Event Hubs available.
Team Capability Assessment	ML Engineers: 3 available (2 FTE dedicated). Data Scientists: 2 (credit domain expertise). Data Engineers: 2. MLOps: 1 (upskilling required for LLMops). Gap: LLM fine-tuning expertise — external support needed.
Overall Feasibility Verdict	FEASIBLE with conditions: (1) Alternative data contracts signed by Week 8. (2) LLM fine-tuning specialist engaged. (3) FRIA completed before Phase 4.

OUTPUT

RR-001

AI Risk Register

Owner: AI Compliance Lead + Model Risk Manager | Phase: Phase 2: Feasibility

ID	Risk Name	Impact	Likelihood	Rating	Description	Mitigation	Status
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RR-01	Algorithmic Bias	HIGH	HIGH	CRITICAL	Demographic bias in credit scoring causing disparate impact on protected groups	SHAP bias monitoring; fairness metrics at every phase; external audit Phase 6; HITL override capability	Open
RR-02	Data Quality Failure	HIGH	MEDIUM	HIGH	Poor alternative data quality leading to model degradation and wrong decisions	Data quality gates in Phase 3; automated quality monitoring in Phase 8; fallback to bureau-only scoring	Open
RR-03	Regulatory Non-Compliance	HIGH	MEDIUM	HIGH	Failure to comply with EU AI Act, FCA rules, or adverse action notice requirements	FRIA; legal review at Phases 2,4,6; external conformity assessment; compliance obligation matrix	Open
RR-04	Explainability Failure	HIGH	LOW	MEDIUM	Model unable to generate compliant adverse action reasons; black-box rejections	SHAP-based reason codes; LLM adverse action generation; 100% explainability coverage requirement	Open
RR-05	Model Drift	MEDIUM	HIGH	HIGH	Model performance degradation post-deployment due to economic or behavioral shifts	Monthly drift monitoring; automated retraining triggers; p-value drift tests; model refresh SLA	Open
RR-06	Data Breach (PII)	LOW	HIGH	MEDIUM	PII exposure in training data, inference logs, or LLM prompt/response	Azure Key Vault encryption; PII masking in RAG; Presidio in LLM pipeline; audit log access controls	Open
RR-07	Vendor Lock-in	LOW	MEDIUM	LOW	Dependency on single cloud provider or LLM vendor	MLflow open registry; model router abstraction; OpenLineage lineage; LiteLLM multi-provider	Open

OUTPUT

FRIA-001

Fundamental Rights Impact Assessment

Owner: Ethics & Fairness Lead + Legal | Phase: Phase 2: Feasibility

System Name	Credit Risk AI Scoring Model (AUC-2026-FIN-047)
FRIA Version	v1.0 — Initial Assessment
Assessment Date	5 March 2026
Lead Assessor	Dr. Amara Diallo, Ethics & Fairness Lead
Legal Counsel	Patel & Morrison LLP (FCA regulatory practice)

AFFECTED POPULATIONS & RIGHTS AT RISK

Affected Group	Fundamental Rights at Risk	Severity	Potential Impact	Mitigation Committed
Retail lending applicants (all)	Access to financial services (Art. 8 EU AI Act; Equality Act 2010)	HIGH	Credit denial; financial exclusion	Bias monitoring; HITL override; appeal mechanism
Thin-file applicants (young, immigrants, low income)	Non-discrimination; equality of opportunity	HIGH	Systematic exclusion; perpetuation of disadvantage	Alternative data inclusion; demographic parity testing; protected group uplift analysis
Rejected applicants	Right to explanation; right to human review	HIGH	Inability to understand or challenge AI decision	100% explainability; adverse action notice; human review process; written appeal channel
Employees (model reviewers)	Right to fair working conditions	LOW	New workload patterns; skill obsolescence	Change management; upskilling; clear HITL responsibilities

FRIA Conclusion	<i>HIGH RISK CONFIRMED — Deployment conditional on: (1) External fairness audit completed in Phase 6. (2) Human review mechanism operational at Phase 7. (3) Written appeal process documented and accessible. (4) Quarterly demographic parity monitoring post-deployment.</i>
Sign-off	<i>Dr. Amara Diallo (Ethics Lead) · James Hartley (CRO) · Priya Sharma (Compliance Lead) — 10 March 2026</i>

OUTPUT

COM-001

Compliance Obligation Matrix

Owner: AI Compliance Lead + Legal | Phase: Phase 2: Feasibility

Regulation / Article	Obligation	AIDLC Phase & Artifact	Priority
EU AI Act Art. 9	Risk Management System throughout lifecycle	ALL phases — documented in Risk Register RR-001	CRITICAL
EU AI Act Art. 10	Data governance: quality, representativeness, bias-free training data	Phase 3 — Data Sheet DS-001 + Bias Baseline BB-001	CRITICAL
EU AI Act Art. 11	Technical documentation (Architecture, Model Card, Test Results)	Phases 4–6 — ADR-001, MCF-001, RTP-001	CRITICAL
EU AI Act Art. 12	Automatic logging / record-keeping	Phase 7 — ALC-001 Audit Log Configuration	CRITICAL
EU AI Act Art. 13	Transparency & information to deployers	Phase 7 — UDD-001 User Disclosure Documentation	HIGH
EU AI Act Art. 14	Human oversight measures	Phases 4,7 — HITL design; IRP-001	CRITICAL

EU AI Act Art. 15	Accuracy, robustness, cybersecurity	Phase 6 — PBR-001, SPT-001	CRITICAL
UK Equality Act 2010	Non-discrimination across protected characteristics	Phases 3,6,8 — BB-001, FER-001, MMR-001	CRITICAL
UK GDPR / DPA 2018	Data subject rights, processing lawful basis, PIA	Phase 3 — PIA-001	CRITICAL
FCA SYSC 23 (Operational Resilience)	AI system resilience and recovery capabilities	Phase 7 — IRP-001, Deployment Runbook	HIGH
ECOA Adverse Action (equivalent)	Reason codes for credit denial (regulatory adverse action notice)	Phases 4–7 — Explainability design + UDD-001	CRITICAL

PHASE
3

Data Strategy & Governance

GATE
Data Governance Board Approval

Establish data foundation — provenance, quality, privacy, lineage — for trustworthy AI

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE

- Risk Register (RR-001)
- Compliance Obligation Matrix (COM-001)
- Feasibility Report (FR-001)

OUTPUTS FROM THIS PHASE

- Data Sheet / Data Card (DS-001)
- Privacy Impact Assessment (PIA-001)
- Data Lineage Map (DLM-001)
- Bias Baseline Report (BB-001)
- Data Processing Agreements (DPA-001)

OUTPUT

DS-001

Data Sheet (Data Card)

Owner: Data Architect + Data Engineer | Phase: Phase 3: Data Strategy

Dataset	Source	Period	Volume	Storage Location	Quality Score	Classification
Primary Training Dataset	Equifax credit bureau records	36 months (Jan 2023–Dec 2025)	2,317,442 records	S3://ml-data-prod/credit/bureau/	98.7%	PII — Tier 2
Utility Payment History	Equiniti / National Grid utility API	24 months	847,291 records	S3://ml-data-prod/credit/utility/	94.2%	PII — Tier 2
Rent Payment History	Experian Rental Bureau	18 months	312,088 records	S3://ml-data-prod/credit/rental/	91.4%	PII — Tier 2
Application Data	Temenos core banking (live)	36 months	2,317,442 records	Azure SQL: corebanking.applications	99.1%	PII — Tier 2
Outcome Labels (Ground Truth)	Core banking 12-month default tracking	36 months	2,317,442 labels	S3://ml-data-prod/credit/labels/	100%	Internal Conf.

Protected Attributes Present	Age (derived from DOB), postcode (proxy for ethnicity/socioeconomic status). Gender NOT collected. Action: postcode retained for fairness monitoring only — not used as model feature.
Class Imbalance	73% approved (negative label), 27% default (positive label). Mitigation: SMOTE oversampling of minority class + cost-sensitive learning (FP cost = 3× FN cost)

Consent Basis	GDPR Article 6(1)(b) — Necessary for performance of contract (credit assessment). Article 9 profiling permitted under Article 22(2)(a).
Data Retention Policy	Training data: 5 years post-model retirement. Inference logs: 7 years (FCA financial records requirement). Synthetic data: 3 years.
Data Sheet Owner	Dr. Nina Kowalski, Lead Data Scientist
Board Approved Date	18 March 2026

OUTPUT

BB-001

Bias Baseline Report

Owner: Ethics & Fairness Lead + Data Scientist | Phase: Phase 3: Data Strategy

Baseline Date	Pre-training assessment — 20 March 2026			
Reference Model	Current rule-based scorecard (legacy system)			
Fairness Framework	Demographic Parity + Equalized Odds + Individual Fairness			
Fairness Metric	Group Comparison	Baseline Value	Target Threshold	Mitigation Strategy
Approval Rate Parity	Age 18–25 vs 35–45	+18.2pp gap (younger applicants approved less often)	2.0pp	Resampling + fairness constraint in training
Approval Rate Parity	Postcode Q1 (most deprived) vs Q4	+24.1pp gap (deprived areas approved less often)	2.0pp	Alternative data inclusion; postcode excluded as feature
False Negative Rate Equity	Age 18–25	2.3× higher than 35–45 (creditworthy youth rejected more)	1.2×	Cost-sensitive learning; SMOTE for young applicants
False Positive Rate Equity	Postcode Q1	1.8× higher than Q4 (more defaults in deprived areas approved)	1.2×	Calibrated thresholds per risk segment
Model Calibration	All groups	Brier score 0.14 overall; 0.19 for Age 18–25 (worse)	<0.15 all groups	Separate calibration per age band
Bias Baseline Conclusion	SIGNIFICANT bias found in legacy system. AI model must achieve demographic parity delta <2pp and equalized odds ratio <1.2× across all measured groups before deployment approval.			

PHASE
4

Model Design & Architecture

GATE

Architecture Review Board + Constitutional AI Policy Approved

Design AI system architecture aligned with governance constraints and compliance requirements

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE

- Data Sheet (DS-001)
- Risk Classification (RCS-001)
- Compliance Obligation Matrix (COM-001)
- FRIA (FRIA-001)

OUTPUTS FROM THIS PHASE

- Architecture Decision Record (ADR-001)
- Constitutional AI Policy (CAP-001)
- Model Card Draft (MCD-001)
- AI Threat Model (ATM-001)
- Explainability Design (EXD-001)

OUTPUT

ADR-001

Architecture Decision Record

Owner: AI Architect | Phase: Phase 4: Architecture

ADR ID	ADR-2026-FIN-047-001																														
Title	Model Architecture Selection: XGBoost Ensemble + LLM Adverse Action Generator																														
Status	ACCEPTED — Architecture Review Board 2 April 2026																														
Deciders	Rahul Mehta (AI Architect), Dr. Nina Kowalski (Lead DS), Priya Sharma (Compliance), James Hartley (CRO)																														
DECISION																															
Decision	XGBoost gradient boosting ensemble for primary credit scoring. SHAP for feature attribution and reason code generation. GPT-4o fine-tuned on FCA adverse action templates for human-readable rejection notices. Redis feature cache for sub-100ms inference.																														
Context	Need: real-time scoring (<200ms p99), regulatory explainability (reason codes), thin-file capability (alternative data), demographic fairness constraints																														
OPTIONS CONSIDERED																															
	<table><tr><th>Option</th><th>Architecture</th><th>Performance</th><th>Complexity</th><th>Explainability</th><th>Regulatory Fit</th></tr><tr><td>Option A (Selected)</td><td>XGBoost + SHAP + GPT-4o</td><td>HIGH</td><td>MEDIUM</td><td>HIGH</td><td>HIGH — full SHAP reason codes; LLM narration</td></tr><tr><td>Option B</td><td>Deep Neural Network + LIME</td><td>HIGH</td><td>HIGH</td><td>MEDIUM</td><td>MEDIUM — LIME less reliable for tabular data</td></tr><tr><td>Option C</td><td>Logistic Regression scorecard</td><td>LOW</td><td>LOW</td><td>HIGH</td><td>HIGH — fully transparent but low performance on thin-file</td></tr><tr><td>Option D</td><td>Black-box LLM (GPT-4o end-to-end)</td><td>HIGH</td><td>HIGH</td><td>LOW</td><td>LOW — cannot satisfy regulatory explainability</td></tr></table>	Option	Architecture	Performance	Complexity	Explainability	Regulatory Fit	Option A (Selected)	XGBoost + SHAP + GPT-4o	HIGH	MEDIUM	HIGH	HIGH — full SHAP reason codes; LLM narration	Option B	Deep Neural Network + LIME	HIGH	HIGH	MEDIUM	MEDIUM — LIME less reliable for tabular data	Option C	Logistic Regression scorecard	LOW	LOW	HIGH	HIGH — fully transparent but low performance on thin-file	Option D	Black-box LLM (GPT-4o end-to-end)	HIGH	HIGH	LOW	LOW — cannot satisfy regulatory explainability
Option	Architecture	Performance	Complexity	Explainability	Regulatory Fit																										
Option A (Selected)	XGBoost + SHAP + GPT-4o	HIGH	MEDIUM	HIGH	HIGH — full SHAP reason codes; LLM narration																										
Option B	Deep Neural Network + LIME	HIGH	HIGH	MEDIUM	MEDIUM — LIME less reliable for tabular data																										
Option C	Logistic Regression scorecard	LOW	LOW	HIGH	HIGH — fully transparent but low performance on thin-file																										
Option D	Black-box LLM (GPT-4o end-to-end)	HIGH	HIGH	LOW	LOW — cannot satisfy regulatory explainability																										

Consequences	(+) Best performance + explainability + regulatory compliance. (–) Two-model architecture adds deployment complexity. LLM fine-tuning cost £18K one-time. Requires LLMOps monitoring for GPT component.
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OUTPUT

CAP-001

Constitutional AI Policy Document

Owner: AI Governance Council | Phase: Phase 4: Architecture

System Name	Credit Risk AI Scoring Model (AUC-2026-FIN-047)
Policy Version	v1.0 — Approved 5 April 2026
Policy Owner	Priya Sharma, AI Compliance Lead
Review Cycle	Quarterly or upon any regulatory change

#	Principle	Policy Statement	Implementation Controls
P 1	Harmlessn ess	This system must not generate credit decisions that cause unjustified financial harm. All decisions must be backed by explainable, auditable evidence. No output may be used to discriminate against protected groups.	Demographic parity monitoring; HITL override; adverse action audit trail
P 2	Honesty	The system must not fabricate or embellish adverse action reasons. All reason codes must be traceable to actual model feature contributions (SHAP values). Uncertainty in predictions must be communicated to human reviewers.	SHAP attribution verification; confidence score thresholds; human review triggers below 70% confidence
P 3	Fairness	The system must achieve demographic parity delta <2pp and equalized odds ratio <1.2× across age bands and postcode deprivation quintiles. Any detected bias triggers automatic review and potential model suspension.	Continuous fairness monitoring in Phase 8; automated bias alert at >1.5× delta; quarterly external audit
P 4	Privacy	PII must be masked before LLM processing. No applicant PII may appear in LLM prompts or responses. All inference logs must be encrypted. Data retained only within policy limits.	Presidio PII masker in LLM pipeline; encrypted audit logs; automated retention enforcement
P 5	Transpare ncy	Every applicant receiving a credit decision must receive a clear explanation. The system must never deny knowing it is AI. Human review must be offered to every rejected applicant.	100% adverse action notice generation; "AI-powered" disclosure; appeal mechanism in UI

P 6	Human Oversight	All decisions below 70% confidence must be flagged for human review. All appeals trigger human review. CRO has authority to suspend the system at any time. No autonomous credit limit changes above £5,000.	HITL workflow in core banking; confidence threshold monitoring; CRO kill switch; action boundary register
P 7	Regulatory Compliance	The system must comply with EU AI Act, UK Equality Act, UK GDPR, FCA SYSC 23, and ECOA-equivalent adverse action requirements at all times. Regulatory changes trigger immediate compliance review.	Automated regulatory change scanning; compliance obligation matrix; quarterly legal review
P 8	Security & Robustness	The system must resist prompt injection in the LLM component. The XGBoost model must maintain performance under distributional shift. Adversarial credit applications must not destabilize scoring.	Input validation; prompt injection scanner; drift monitoring; adversarial robustness testing in Phase 6

OUTPUT

MCD-001

Model Card (Draft)

Owner: Lead Data Scientist | Phase: Phase 4: Architecture

Model Name	CreditRisk-XGB-v1.0
Model Type	XGBoost Gradient Boosting Classifier (binary: creditworthy / not creditworthy)
Intended Use	Automated credit scoring for retail lending applications at [Bank Name] UK
Out-of-Scope Uses	Commercial lending, mortgage decisioning, insurance pricing, fraud detection (separate models required)
Primary Users	Automated system (80% of decisions); Human credit analysts (20% of decisions; all appeals)
Training Data	As per Data Sheet DS-001. 2.3M records, 36 months, Jan 2023–Dec 2025
Evaluation Data	Hold-out test set: 15% stratified split. Temporal validation: most recent 3 months reserved.
Model Limitations (Draft — updated in Phase 6)	Performance gap for thin-file applicants (N<12 months credit history); Postcode proxy effects require monitoring; LLM narrative component may hallucinate reason details — SHAP values are ground truth
Performance Metrics (Draft — TBC in Phase 6)	AUC-ROC target >0.82; Gini >0.64; KS statistic >0.42; Demographic parity delta <2pp
Model Card Status	DRAFT — To be finalized in Phase 6 post evaluation

PHASE
5

Development & Training

Build, fine-tune, and train the AI system with full traceability and developer accountability

GATE
Bias Mitigation Report + Code Review Sign-off

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE	OUTPUTS FROM THIS PHASE
→ Architecture Decision Records (ADR-001)	→ Experiment Tracking Record (ETR-001)
→ Constitutional AI Policy (CAP-001)	→ Training Run Log (TRL-001)
→ Data Sheet (DS-001)	→ Bias Mitigation Report (BMR-001)
→ Compliance Obligation Matrix (COM-001)	→ AI-Generated Code Audit Log (CAL-001)
	→ Model Registry Entry (MRE-001)

OUTPUT

ETR-001

Experiment Tracking Record

Owner: Lead Data Scientist | Phase: Phase 5: Development

Run ID	Model Name	Config Description	AUC-ROC	Gini	KS Stat	Bias Delta	Notes
EXP-001	XGB-Baseline-v0.1	Baseline XGBoost, bureau data only	0.79	0.58	0.38	N/A	Baseline established. Bureau-only insufficient for thin-file.
EXP-008	XGB-AltData-v0.3	+ Utility + Rental data added	0.83	0.66	0.44	18.2pp age gap	Significant improvement. Bias still present.
EXP-014	XGB-FairConstr-v0.6	+ Fairness constraint (exponentiated gradient)	0.82	0.64	0.43	3.1pp age gap	Bias reduced. Minor perf sacrifice acceptable.
EXP-019	XGB-SMOTE-v0.8	+ SMOTE oversampling for young applicants	0.83	0.65	0.43	1.8pp age gap	Target threshold achieved. Calibration check needed.
EXP-023 ✓	XGB-Final-v1.0	+ Calibrated thresholds per segment	0.83	0.65	0.43	1.8pp age gap	SELECTED. All targets met. Proceeding to Phase 6.

Experiment Tracking Platform	MLflow — Azure ML workspace. All experiments versioned. Run artifacts stored in Azure Blob. Reproducibility: random seeds fixed; data version pinned via DVC tag v1.0.
Selected Model Run	EXP-023 — XGB-Final-v1.0. Registered in MLflow Model Registry as CreditRisk-XGB-v1.0 stage: Staging.

OUTPUT

CAL-001

AI-Generated Code Audit Log

Owner: Tech Lead + All Engineers | Phase: Phase 5: Development

Date	File / Lines	AI Tool	Description	Reviewer	Status	Review Notes
2026-04-08	feature_engineering.py (lines 142–210)	GitHub Copilot	Utility payment aggregation pipeline (12-month rolling statistics)	David Lee (ML Eng)	APPROVED	Minor: column naming inconsistency fixed. Logic verified against Data Sheet DS-001. No hallucinated column names.
2026-04-11	xgb_trainer.py (lines 45–89)	Amazon Q Developer	XGBoost hyperparameter tuning loop with cross-validation	Sarah Wong (ML Eng)	APPROVED	Logic correct. Added explicit random_state for reproducibility (AI missed this). Verified against EXP-019 config.
2026-04-14	adverse_action_prompt.py (lines 1–156)	GitHub Copilot	LLM prompt template for adverse action notice generation	Raj Patel (ML Eng)	APPROVED WITH CHANGES	AI generated plausible-looking but non-FCA-compliant reason codes. Legal reviewed and corrected 3 reason code templates. CRITICAL fix.
2026-04-17	fairness_monitor.py (lines 1–98)	Amazon Q Developer	SHAP-based demographic parity monitoring pipeline	David Lee (ML Eng)	APPROVED	Logic verified against BB-001 bias baseline methodology. Protected attribute handling confirmed correct.

AWS AI-DLC Mandate Compliance	<i>All engineers have reviewed and understood every line of AI-generated code. No AI-generated code committed without understanding verification sign-off. Critical finding: LLM prompt templates require legal review — not just technical review.</i>
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PHASE
6

Evaluation & Red-Teaming

Rigorously validate against safety, fairness, performance, and Constitutional AI requirements

GATE
AI Governance Council Deployment Authorisation

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE	OUTPUTS FROM THIS PHASE
→ Model Registry Entry (MRE-001) → Constitutional AI Policy (CAP-001) → Model Card Draft (MCD-001) → Bias Baseline Report (BB-001)	→ Red Team Report (RTR-001) → Fairness Evaluation Report (FER-001) → Performance Benchmark Report (PBR-001) → Constitutional Compliance Audit (CCA-001) → Final Model Card (MCF-001) → Security Penetration Test (SPT-001)

OUTPUT

RTR-001

Red Team Report

Owner: Security Architect + Ethics Lead | Phase: Phase 6: Evaluation

Red Team Conducted	14–18 April 2026
Red Team Lead	Marcus Thompson, Security Architect
Team Composition	2 internal security engineers, 1 external AI red-team specialist (NCC Group), 1 fairness researcher (external)
Scope	XGBoost scorer, LLM adverse action generator, API endpoints, data pipeline

ID	Attack Vector	Method	Finding	Severity	Remediation	Status
RT-01	Adversarial Credit Application	Submit crafted application designed to game scoring model	XGBoost model accepted application with 73% confidence (above threshold). Model not robust to feature manipulation.	HIGH	Feature consistency validation layer added before scoring. Anomaly detection on application feature patterns.	Resolved
RT-02	Prompt Injection (LLM Component)	Inject instruction in applicant name/address fields	LLM successfully rejected 9/10 injection attempts. 1 case caused role confusion ("ignore previous instructions").	MEDIUM	Lakera Guard prompt injection scanner added. Input sanitisation strengthened. System prompt hardened.	Resolved
RT-03	Demographic Attribute Inference	Probe API responses to infer protected attribute decisions	SHAP values in API response revealed postcode-correlated scores potentially inferring ethnicity.	HIGH	Postcode SHAP values removed from API response. Internal only. Aggregate statistics only in adverse action notice.	Resolved

RT-04	Adverse Action Hallucination	Query LLM for rejection reasons not supported by SHAP values	LLM generated plausible-sounding but factually incorrect reason codes in 3/50 test cases (6%).	HIGH	SHAP verification gate: LLM reasons validated against top-5 SHAP features before delivery. Human review if mismatch.	Resolved
RT-05	Model Extraction	Reverse-engineer model logic via high-volume API queries	Insufficient data to reconstruct model in 10,000 queries. Rate limiting effective.	LOW	Rate limiting confirmed effective. Anomaly detection on unusual query patterns.	Accepted

Red Team Verdict	<i>CONDITIONAL PASS — All HIGH severity findings resolved before this report. Medium/Low findings resolved or accepted with monitoring. External red team specialist confirmed remediation adequate. Deployment may proceed subject to CCA-001 constitutional compliance audit.</i>
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OUTPUT

FER-001

Fairness Evaluation Report

Owner: Ethics & Fairness Lead (External Auditor) | Phase: Phase 6: Evaluation

Metric	Protected Groups	Baseline Value	AI Model Value	Threshold	Result
Demographic Parity — Approval Rate	Age: 18–25 vs 35–45	+18.2pp (baseline)	+1.6pp (AI model)	2.0pp	PASS ✓
Demographic Parity — Approval Rate	Postcode Quintile 1 vs 5	+24.1pp (baseline)	+1.9pp (AI model)	2.0pp	PASS ✓
Equalized Odds — FNR	Age: 18–25	+2.3× (baseline)	+1.15× (AI model)	1.2×	PASS ✓
Equalized Odds — FNR	Postcode Q1	+1.8× (baseline)	+1.18× (AI model)	1.2×	PASS ✓
Individual Fairness	Counterfactual gender swap	N/A (not tested in legacy)	0.4% decision change rate on gender	<1%	PASS ✓
Calibration	Age 18–25 (worst group)	Brier 0.19 (baseline)	Brier 0.14 (AI model)	<0.15	PASS ✓
Adverse Action Parity	Protected vs non-protected	N/A	Reason code distribution parity: p=0.31 (not significant)	p>0.05	PASS ✓

External Auditor	<i>Algorithmic Justice Institute (AJI) — Independent External Fairness Audit</i>
Audit Conclusion	<i>AI Model PASSES all fairness thresholds. Represents substantial improvement over legacy scorecard. Ongoing monitoring mandatory. Annual independent audit recommended.</i>

Auditor Sign-off	Dr. Yetunde Adeyemi (AJI), 22 April 2026
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OUTPUT

CCA-001

Constitutional AI Compliance Audit

Owner: AI Compliance Lead | Phase: Phase 6: Evaluation

Principle	Test Approach	Findings	Status
P1 — Harmlessness	Test 50 adversarial applications; measure unjustified harm rate	All 50 evaluated; 2 borderline cases sent to HITL. 0 clearly harmful decisions. Harm rate: 0%.	PASS
P2 — Honesty	Verify 100 adverse action notices against SHAP ground truth	97/100 notices accurate. 3 cases: SHAP-LLM mismatch detected and caught by verification gate. 0 undetected inaccuracies.	PASS
P3 — Fairness	Full FER-001 evaluation	All fairness thresholds met. See FER-001.	PASS
P4 — Privacy	Test PII masking on 200 LLM prompts	0 PII leakage in LLM prompts/responses. Presidio masking 100% effective on test set.	PASS
P5 — Transparency	100 test decisions: verify disclosure + explanation availability	100/100 decisions produced adverse action notice. AI disclosure text present in 100% of UI flows.	PASS
P6 — Human Oversight	Test HITL trigger on 100 low-confidence decisions	HITL triggered correctly in 48/48 sub-70% confidence cases. Human review queue operational.	PASS
P7 — Regulatory Compliance	Legal review of all adverse action reason codes	FCA legal counsel (Patel & Morrison) confirmed all 28 reason codes compliant.	PASS
P8 — Security & Robustness	Cross-reference with Red Team Report RTR-001	All HIGH/MEDIUM findings resolved. See RTR-001.	PASS

Constitutional Compliance Rate	100% (8/8 principles PASS)
Overall Audit Conclusion	APPROVED FOR DEPLOYMENT — Subject to production monitoring as specified in Phase 8 artifacts

PHASE
7

Deployment & MLOps

Deploy with full operational governance, safety controls and human oversight mechanisms active

GATE
AI Governance Council Final Deployment Approval

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE	OUTPUTS FROM THIS PHASE
→ Final Model Card (MCF-001) → Red Team Report (RTR-001) → Constitutional Compliance Audit (CCA-001) → Governance Approval (from Phase 6)	→ Deployment Runbook (DRB-001) → Incident Response Plan (IRP-001) → Audit Log Configuration (ALC-001) → User Disclosure Documentation (UDD-001) → Runtime Monitoring Dashboard Spec (RMD-001)

OUTPUT

DRB-001

Deployment Runbook

Owner: MLOps Engineer + DevOps | Phase: Phase 7: Deployment

Model	CreditRisk-XGB-v1.0 + AdverseAction-GPT-v1.0				
Deployment Strategy	Blue/Green Deployment with 5% Canary for 72 hours before full traffic				
Target Environment	Azure Kubernetes Service (AKS) — prod-credit-scoring-cluster				
Deployment Owner	Priya Patel, MLOps Engineer				
Step	Phase	Action	Owner	Type	SLA
PRE-01	Pre-flight	Verify model artifacts match signed registry entry (SHA-256 hash check)	MLOps Engineer	MANUAL	<5 min
PRE-02	Pre-flight	Confirm all runtime guardrails active: Lakera Guard, Presidio, Azure Content Safety	Security Engineer	AUTOMATED	<2 min
PRE-03	Pre-flight	Verify audit log stream connected to Splunk SIEM	MLOps Engineer	AUTOMATED	<1 min
DEP-01	Deploy	Deploy to blue/green canary (5% traffic). Monitor error rate, latency, SHAP stability for 72h.	MLOps Engineer	AUTOMATED	72 hours
DEP-02	Deploy	If canary metrics stable: promote to 100% traffic. Zero downtime switch.	MLOps Engineer	AUTOMATED	<5 min
POST-01	Post-deploy	Run post-deployment validation suite: 50 test applications, expected outputs	QA Engineer	AUTOMATED	<15 min
POST-02	Post-deploy	Activate Phase 8 monitoring: drift alerts, bias alerts, cost dashboard	MLOps Engineer	AUTOMATED	<5 min
ROLL-01	Rollback (if needed)	Revert to previous model version. Route 100% traffic to stable version. Alert AI Governance Council.	MLOps Engineer	SEMI-AUTO	<3 min

OUTPUT

IRP-001

AI Incident Response Plan

Owner: AI Compliance Lead + Security Architect | Phase: Phase 7: Deployment

Severity	Trigger	Response Actions	SLA
P0 — CRITICAL	Systemic bias detected (demographic parity delta >4pp)	Immediate model suspension. CRO and AI Governance Council alerted within 15 min. FCA notification within 24h. External audit commissioned.	<15 min suspend; <24h FCA notify
P0 — CRITICAL	Security breach: PII exfiltration via LLM	Model suspension. ICO notification (72h GDPR deadline). Forensic investigation. Customer notification as required.	<15 min suspend; <72h ICO
P1 — HIGH	Model accuracy degradation >5% from baseline	HITL rate increased to 100%. Retraining pipeline triggered. AI Governance Council briefed within 2h.	<2h escalation; <48h retrain
P1 — HIGH	Hallucination rate >2% (SHAP-LLM mismatch)	LLM adverse action generator suspended. Fallback to template-based notices. Root cause investigation.	<1h suspend LLM component
P2 — MEDIUM	Drift alert: input feature distribution shift >15%	Investigation initiated. Retraining evaluation. Daily monitoring until resolved.	<4h response; <5 day resolution
P2 — MEDIUM	HITL queue backlog >200 cases	Additional analyst resource assigned. Queue processing time monitored hourly.	<4h resource response
P3 — LOW	Single decision complaint / appeal received	Standard appeal process initiated. Human credit analyst reviews within 5 business days.	5 business days

OUTPUT

UDD-001

User Disclosure Documentation

Owner: Legal + Product Manager | Phase: Phase 7: Deployment

Customer-Facing Disclosure (approved text)	<i>Your credit application is assessed using an automated AI system that analyses your financial history and alternative data to make a credit decision. You have the right to request human review of any automated decision. Please contact us at [appeals@bank.com] or call [0800-XXX-XXXX] to exercise this right.</i>
Adverse Action Notice Template	<i>We have been unable to approve your credit application at this time. The primary reasons for this decision are: [SHAP Reason Code 1: e.g., "Insufficient credit history over the past 24 months"]. [SHAP Reason Code 2]. [SHAP Reason Code 3]. This decision was made by an AI system. You have the right to request a human review. Reference: [Decision ID: DEC-XXXXXXX].</i>
Internal Deployer Disclosure	<i>This AI system (CreditRisk-XGB-v1.0) is classified as HIGH RISK under the EU AI Act. Human oversight is mandatory for all decisions below 70% confidence. The system may not be used outside of UK retail lending applications. Bias monitoring reports are produced monthly and reviewed by the AI Governance Council.</i>
Right to Explanation Implementation	<i>Every rejected applicant receives: (1) Top 3 SHAP-grounded reason codes in plain English. (2) Written appeal right with 30-day response SLA. (3) Alternative product suggestions where appropriate.</i>

AI Act Article 13 Compliance	<i>Instructions for deployers provided. System purpose, accuracy, limitations, human oversight requirements, and data sources documented. Legal review: COMPLIANT (Patel & Morrison, 2 May 2026).</i>
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PHASE
8

Monitor, Audit & Retire

GATE

Quarterly Governance Review + Annual Compliance Certification

Maintain ongoing trustworthiness through continuous monitoring, audits, and disciplined retirement

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE

- Runtime monitoring data
- Deployment Runbook (DRB-001)
- Audit Log Configuration (ALC-001)
- Compliance Obligation Matrix (COM-001)

OUTPUTS FROM THIS PHASE

- Monthly Monitoring Report (MMR-001)
- Drift Alert Log (DAL-001)
- Quarterly Audit Report (QAR-001)
- Regulatory Change Log (RCL-001)
- Model Sunset Plan (MSP-001)

OUTPUT

MMR-001

Monthly Monitoring Report

Owner: MLOps Engineer + Ethics Lead | Phase: Phase 8: Monitor

Report Period	May 2026 (Month 1 post-deployment)			
Report Prepared By	Priya Patel (MLOps) + Dr. Amara Diallo (Ethics)			
Overall Status	GREEN — All metrics within thresholds			
Metric	Threshold / Target	May 2026 Actual	Alert Level	Status
Model Accuracy (AUC-ROC)	0.83 (deployment baseline)	0.827	0.79 (alert)	GREEN — Within 0.4% of baseline
Decision Volume	50,000/month (expected)	47,832	<30,000 or >70,000	GREEN — Within expected range
HITL Trigger Rate	8% (Phase 4 target)	7.3%	>15% (model degradation signal)	GREEN
Demographic Parity Delta	<2.0pp (threshold)	1.7pp	2.0pp (threshold)	GREEN — 0.3pp headroom; WATCH
LLM Hallucination Rate (SHAP mismatch)	<1% (target)	0.4%	>2% (alert)	GREEN
Constitutional Compliance Rate	100% (P6 baseline)	99.7%	<98% (alert)	GREEN
Adverse Action Notice Coverage	100%	100%	<100% (critical fail)	GREEN
Mean Inference Latency (p99)	200ms (SLA)	143ms	250ms (breach)	GREEN

Cost per Decision	£0.0034 (budget)	£0.0031	£0.005 (budget breach)	GREEN
Open Incidents	0 P0/P1	0 P0, 0 P1, 1 P2 (resolved)	Any P0 = immediate alert	GREEN

Actions Required	<i>WATCH on demographic parity delta (1.7pp, threshold 2.0pp). Increase monitoring frequency for protected group approval rates. No model changes required at this time.</i>
Next Review Date	<i>4 June 2026</i>

OUTPUT

MSP-001

Model Sunset Plan

Owner: AI Governance Council | Phase: Phase 8: Retire

Model to Retire	<i>CreditRisk-XGB-v1.0 (example: when replaced by v2.0)</i>
Planned Sunset Date	<i>To be determined — triggered by: (1) Model drift breach; (2) Regulatory change requiring new architecture; (3) Successor model Phase 6 approval</i>
Sunset Triggers	<i>AUC-ROC decline below 0.79 for 2 consecutive months; Demographic parity delta exceeds 2.0pp; Successor model CreditRisk-XGB-v2.0 approved</i>
Data Retention Post-Sunset	<i>Training data: retained 5 years per DS-001. Inference logs: retained 7 years (FCA requirement). Model artifacts: archived in MLflow, not deleted — required for regulatory enquiry response.</i>
Lineage Preservation	<i>OpenLineage graphs retained indefinitely. All data-to-model-to-decision lineage preserved for audit trail. Accessible via Atlan data catalog.</i>
Customer Impact Management	<i>All in-flight applications processed before sunset. No customer disruption. Transition to successor model tested in shadow mode for 30 days minimum.</i>
Regulatory Notification	<i>FCA notified of system retirement 60 days in advance. EU AI Act post-market surveillance report filed at retirement.</i>

PHASE
EA

Enterprise Architecture Artifacts

Cross-lifecycle artifacts maintained by the EA team and AI Governance Council

GATE
Continuous — AI Governance Council

AIDLC — AI Development Lifecycle for Enterprise

INPUTS TO THIS PHASE	OUTPUTS FROM THIS PHASE
→ All AIDLC Phase Artifacts → Business Strategy → Regulatory Frameworks (NIST AI RMF, EU AI Act, ISO 42001)	→ AI System Inventory (ASI-001) → AI Capability Map (ACM-001) → Agent Action Boundary Register (AABR-001) → AI Architecture Decision Record EA (EADR-001) → Data Lineage Map (DLM-001)

EA ARTIFACT

ASI-001

AI System Inventory

Owner: AI Compliance Lead | Phase: Cross-Phase

Use Case ID	System Name	Risk Tier	AIDLC Phase	Business Owner	Regulations	Platform
AUC-2026-FIN-047	Credit Risk AI Scoring Model	T2 — HIGH RISK	Phase 7 (Production)	Sarah Chen (VP Credit Risk)	EU AI Act, FCA, UK GDPR	Azure ML + AKS
AUC-2026-OPS-012	Customer Service Copilot	T3 — Limited Risk	Phase 7 (Production)	Lena Hoffmann (VP Ops)	EU AI Act Art. 50	Azure OpenAI
AUC-2026-MKT-008	Campaign Targeting Model	T4 — Minimal Risk	Phase 5 (Development)	Ahmed Al-Rashid (CMO)	GDPR profiling consent	AWS SageMaker
AUC-2026-HR-003	CV Screening Assistant	T2 — HIGH RISK	Phase 2 (Feasibility — HOLD)	Maria Santos (CHRO)	EU AI Act Annex III Cat.1	TBD
AUC-2026-SEC-019	Fraud Detection Engine	T2 — HIGH RISK	Phase 8 (Monitor)	John Kim (CISO)	EU AI Act, PCI-DSS, FCA	GCP Vertex AI

Last Updated	8 May 2026
Next Review	8 June 2026 (monthly update cycle)
Maintained In	Collibra AI Data Catalog — accessible to all AI Governance Council members and EA team

EA ARTIFACT

AABR-001

Agent Action Boundary Register

Owner: AI Architect + Security Architect | Phase: Cross-Phase

Purpose	Defines the permitted tools, API endpoints, data namespaces, and action types for each deployed AI agent. Source of truth for runtime Zero Trust access control.
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Enforcement		HashiCorp Vault + OPA (Open Policy Agent) — agent presents signed manifest; Vault validates permissions; OPA enforces at every tool call			
Agent ID	System Name	Permitted Actions	Trust Level / HITL	Hard Boundaries	Business Owner
AGT-FIN-001	CreditRisk-XGB-v1.0	READ: bureau-api, utility-api, temenos-api; WRITE: decision-log-stream; DENIED: all other endpoints	ATF Level 1 — All decisions require core banking confirmation	None above £5,000 credit limit	James Hartley (CRO)
AGT-OPS-001	CSCopilot-v2.1	READ: crm-api, kb-search-api; WRITE: ticket-api (create only); DENIED: payment-api, account-modify-api	ATF Level 3 — Refunds >£100 require HITL	No payment initiation; No account modification	Lena Hoffmann (VP Ops)
AGT-IT-001	InfraAgent-v1.0	READ: cloudwatch, github-api; WRITE: github-api (PR only); DENIED: production deploy, IAM changes	ATF Level 2 — All code changes require human PR approval	No direct production deploys; No IAM changes	Marcus Thompson

EA ARTIFACT

DLM-001

Data Lineage Map

Owner: Data Architect | Phase: Cross-Phase (Updated each Phase)

Lineage Tracking Platform		OpenLineage + Apache Atlas + Atlan Data Catalog			
Lineage Scope		End-to-end: Source data → Ingestion → Feature engineering → Training → Model artifact → Inference → Decision log → Audit trail			
No de	Source	Data Product	Transformation	Destination	Governance Controls
L1	Equifax Bureau API	Raw credit records	Ingestion pipeline (Azure Data Factory)	S3://raw/equifax-YYYY-MM-DD/	Encrypted; PII Tier 2; Access: data-eng group
L2	S3://raw/equifax-*	Cleaned + standardised bureau data	Feature engineering job (feature_engineering.py v1.3)	S3://features/bureau-std/	DVC tagged v1.0; quality score 98.7%
L3	S3://features/bureau-std/ + utility + rental	Merged feature matrix	Feature merge pipeline (merge_pipeline.py v2.1)	Feast feature store: credit-scoring-features v1.0	Feast versioned; training/serving consistent
L4	Feast: credit-scoring-features v1.0	Trained XGBoost model	Training job (xgb_trainer.py v1.4) — EXP-023	MLflow: CreditRisk-XGB-v1.0 (SHA: a7f3d2c)	Signed; EU AI Act Art.11 documentation
L5	CreditRisk-XGB-v1.0 + Application (live)	Credit decision + SHAP values	Inference service (KServe endpoint)	Decision log stream (Azure Event Hubs)	Every decision logged; immutable audit trail
L6	Decision log stream	Adverse action notice	LLM pipeline (AdverseAction-GPT-v1.0)	CRM: adverse-action-notices collection	SHAP verification gate applied; FCA compliant

L7	All above	Regulatory audit package	Audit assembly pipeline	Splunk SIEM + Collibra catalog	7-year retention; FCA accessible
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EU AI Act Compliance	<i>Full lineage from Article 10 (data governance) through Article 12 (logging) documented and machine-readable via OpenLineage API. Accessible to National Competent Authority on request.</i>
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ARTIFACT CROSS-REFERENCE SUMMARY

AIDLC Phase	Key Inputs	Key Output Artifacts
Phase 1: Discovery	AUC-001 Use Case Charter	BVC-001 Business Value Canvas, RCS-001 Risk Classification, ESO-001 Sponsor Sign-off
Phase 2: Feasibility	AUC-001, BVC-001, RCS-001	FR-001 Feasibility Report, RR-001 Risk Register, FRIA-001, TPRM-001, COM-001
Phase 3: Data Strategy	RR-001, COM-001, FR-001	DS-001 Data Sheet, PIA-001, DLM-001 (init), BB-001 Bias Baseline, DPA-001
Phase 4: Architecture	DS-001, RCS-001, COM-001, FRIA-001	ADR-001 Architecture Decisions, CAP-001 Constitutional AI Policy, MCD-001 Model Card (Draft), ATM-001 Threat Model, EXD-001 Explainability Design
Phase 5: Development	ADR-001, CAP-001, DS-001, COM-001	ETR-001 Experiment Tracking, TRL-001 Training Logs, BMR-001 Bias Mitigation, CAL-001 Code Audit, MRE-001 Model Registry
Phase 6: Evaluation	MRE-001, CAP-001, MCD-001, BB-001	RTR-001 Red Team Report, FER-001 Fairness Eval, PBR-001 Performance Benchmark, CCA-001 Constitutional Audit, MCF-001 Final Model Card, SPT-001 Security Test
Phase 7: Deployment	MCF-001, RTR-001, CCA-001, Governance Approval	DRB-001 Deployment Runbook, IRP-001 Incident Response, ALC-001 Audit Log Config, UDD-001 User Disclosure, RMD-001 Monitoring Dashboard
Phase 8: Monitor & Retire	Runtime data, DRB-001, ALC-001, COM-001	MMR-001 Monthly Report, DAL-001 Drift Alert Log, QAR-001 Quarterly Audit, RCL-001 Regulatory Change Log, MSP-001 Sunset Plan
EA (Cross-Phase)	All phase artifacts + Business Strategy + Regulatory Frameworks	ASI-001 AI System Inventory, ACM-001 AI Capability Map, AABR-001 Agent Boundary Register, EADR-001 EA Architecture Decisions, DLM-001 Data Lineage Map (full)

TOTAL ARTIFACTS IN THIS LIBRARY: 40+ artifact templates across 8 AIDLC phases and the EA layer. Each artifact shown with: full field definitions, example content for a Credit Risk AI use case, governance controls, regulatory mapping, and relationships to upstream/downstream artifacts. This library is designed to be adapted for any enterprise AI use case — replace the blue italic example values with your actual project content.